

RFI on the Development of an AI Plan



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Submitted by email to 

Re: Request for Information (RFI) on the Development of an Artificial Intelligence (AI) Action Plan ("Plan")

Introduction

Today, America stands at a critical inflection point in the global artificial intelligence race. As directed by Executive Order 14179 (Removing Barriers to American Leadership in Artificial Intelligence), establishing an effective AI Action Plan requires leveraging the complementary strengths of both public and private sectors while acknowledging current market dynamics. The private sector has established a significant lead in AI development capabilities that government efforts should strategically harness, rather than duplicate.

WindBorne Systems proposes a framework that positions the Federal Government as the world's most sophisticated AI customer while driving American technological leadership through four core principles:

1. **Strategic Resource Allocation:** The Federal Government should redirect resources from internal AI model development toward streamlined procurement pathways for private sector AI solutions, closing capability gaps through strategic acquisition rather than parallel development.
2. **Performance-Based Evaluation:** All Government AI procurement must require rigorous live product demonstrations with standardized evaluation protocols, ensuring taxpayer resources support genuinely effective technologies that exist today, rather than lengthy, duplicative development efforts.
3. **Transformational Efficiency Metrics:** Government implementation of AI should be evaluated based on measurable efficiency multipliers, recognizing that properly deployed AI can deliver order-of-magnitude improvements beyond incremental gains.
4. **Establishing Required Infrastructure:** AI solution implementations will have massive compute requirements beyond the already extensive technical infrastructure supporting existing programs. Used effectively, AI advancements that enable efficiencies in execution can also enable efficiencies in operation.

By adopting these principles, the AI Action Plan can establish a procurement ecosystem that accelerates America's AI advantage, provides superior capabilities to government agencies, and ensures U.S. leadership in the broader technological revolution.

About WindBorne Systems

WindBorne Systems, a Palo Alto-based startup, is innovating both in-situ weather data collection and AI model forecasting to close existing gaps in observation data in the interest of increasing forecast accuracy and prediction efficiency. WindBorne operates WeatherMesh, the world's leading AI-based weather forecasting model—based on published benchmarks worldwide¹—paired with our AI-enabled, long-duration Global Sounding Balloons (GSBs). Together, these innovations represent the cutting edge of atmospheric science and machine learning integration. These systems stand for rigorous demonstration and can deliver immediate efficiency gains to Government weather forecasting capabilities while simultaneously reducing the massive computing costs typically associated with highly performant weather modeling.

¹ WeatherMesh Benchmarks. (2025). <https://windbornesystems.com/forecasts/benchmarks>.

Current AI Landscape and America's Competitive Position

The global AI landscape has fundamentally transformed over the past five years, creating new competitive dynamics between nations and redefining public-private sector relationships. America's private sector leads in foundational model development, with companies investing tens of billions in research infrastructure and talent acquisition. This investment substantially exceeds public sector capabilities, with leading AI firms deploying computational resources that surpass those available to most Government agencies by orders of magnitude.

The economics of AI development have reached a critical threshold: expertise, data, and computational capacity have consolidated within the private sector to a degree that makes Government-led frontier model development increasingly challenging from both resource and talent perspectives. Unlike previous technological revolutions where Government-led research preceded commercial applications, the innovation flow in AI has significantly evolved within the commercial sector over the past decade while Government agencies have had slower rates of development and adoption, or in some cases none at all.

Global Competition: America's Strategic Opportunity

While American commercial companies lead the world in core AI innovation, governments in both China and the European Union have implemented approaches that challenge our nation's long-term advantage.

China has integrated AI development into its national strategy, fostering tight coordination between state entities and technology companies through favorable regulatory environments and data access policies. Rather than developing government-exclusive models, China has created symbiotic relationships where government needs drive private sector development, accelerating deployment across strategic sectors. The European Union, despite regulatory caution evidenced in the AI Act, has pivoted toward selective procurement and adaptation of private sector models for public use. European governments increasingly focus on creating interoperability standards and regulatory frameworks that allow government entities to leverage private innovation while maintaining European values and governance. These contrasting

international approaches demonstrate that effective national AI strategies recognize the complementary roles of public and private sectors rather than attempting to duplicate capabilities across both domains.

The United States faces a profound strategic choice: continue pursuing increasingly resource-intensive Government-led AI development or implement procurement strategies that harness private sector innovation while ensuring Government requirements are met. Evidence strongly suggests the latter approach offers the most viable path to maintaining American leadership. The U.S. is already behind and the luxury of time to develop bespoke solutions does not exist. Fortunately, the need for bespoke solutions can be comprehensively met with commercial solutions that are operational today.

This does not diminish Government's critical role but represents an opportunity to refocus resources where public sector involvement remains essential:

- Establishing comprehensive evaluation frameworks
- Ensuring robust security standards
- Facilitating responsible innovation
- Directing private sector development toward national priorities through strategic procurement

The most successful international models demonstrate that governments achieve maximum impact by establishing clear requirements, creating favorable development environments, and implementing performance-based procurement processes that drive private innovation toward public needs. By shifting from development to strategic acquisition, the Government can leverage America's private sector advantage as a national asset to power continued American dominance worldwide.

Framework for Government AI Procurement

The Federal Government's approach to AI acquisition requires fundamental restructuring to capitalize on America's private sector innovation advantage. Current procurement processes—designed for traditional software and hardware acquisition—are misaligned with the rapid development cycles and unique evaluation requirements of advanced AI systems. We propose a comprehensive framework that prioritizes capability validation, deployment speed, and strategic value creation.

From Parallel Development to Strategic Acquisition

Federal agencies should position themselves as sophisticated customers of America's world-leading AI ecosystem rather than attempting to replicate capabilities internally. This approach recognizes that the most advanced AI capabilities will emerge from private sector innovation, with Government agencies focusing on effectively directing and harnessing these capabilities toward public objectives.

Our recommended procurement framework is organized around three principles:

1. Capabilities Validation Through Demonstration

The gap between claimed and actual AI capabilities necessitates a demonstration-centered evaluation approach. Unlike traditional software where specifications can be verified through documentation review, AI systems require practical performance testing in realistic environments. Federal procurement should require:

- Live capability demonstrations in environments that accurately represent actual Government use cases, with standardized evaluation protocols specific to each application domain
- Extended evaluation periods (4-8 weeks) for mission-critical systems to assess performance stability and adaptation to real-world data variations
- Benchmark comparison against current Government processes to quantify efficiency improvements
- Blind adversarial testing to verify system performance against unanticipated inputs or deliberate manipulation attempts

By centering evaluation on demonstrated rather than documented capabilities, agencies can make procurement decisions based on actual performance rather than theoretical specifications.

2. Accelerated Acquisition Pathways

Current Federal acquisition processes operate too slowly for the AI development cycle. By the time traditional procurement procedures complete, innovative underlying technologies will have undergone multiple iterations, meaning the version actually procured is stale on day one.

To address this misalignment, we recommend:

- Expanding Other Transaction Authority (OTA) usage across civilian agencies for AI procurement, following success within the Department of Defense
- Creating AI-specific Federal Supply Schedule categories with pre-negotiated terms and simplified ordering procedures
- Establishing Government-wide AI Blanket Purchase Agreements (BPAs) that pre-qualify vendors while allowing flexible deployment options
- Implementing phased procurement approaches that begin with rapid pilot deployments before scaling to enterprise implementation
- Raising simplified acquisition thresholds specifically for AI technologies to enable faster purchasing decisions

These mechanisms would allow agencies to move from initial AI solution identification to operational deployment in months rather than years, aligning government procurement timelines with the pace of AI advancement.

3. Values-Based Contracting Models

Traditional Government IT contracts focus primarily on cost and compliance with predetermined specifications. This approach fundamentally misaligns incentives for AI procurement, where value derives from performance improvements rather than feature checklists. The result is that often the most effective technical solutions are disqualified from procurement because of overly prescriptive requirements laid forth by those who are experts in the technological gaps needing to be filled rather than the possible solutions that can fill them. To facilitate the Federal Government's procurement of the most effective, efficient, and advanced solutions, we suggest:

- Outcomes-based contracting that ties compensation to measurable improvements in Government operations (e.g., processing time reduction, accuracy improvements, resource savings)
- Shared savings models where vendors receive a percentage of verified cost reductions achieved through AI implementation
- Scale-based pricing that reduces per-transaction costs as system utilization increases, incentivizing both Government adoption and vendor performance
- Performance incentive payments linked to exceeding baseline efficiency requirements

It is critical to realign incentives, encouraging vendors to maximize Government operational improvements rather than merely delivering compliant systems.

Institutional Infrastructure for Effective Implementation

Successfully implementing this framework requires new institutional capabilities and structures within the Federal Government:

- Establishing AI Solutions Offices within Federal agencies to provide technical expertise and contract vehicles relevant to implementing the Federal AI Action Plan
- Creating agency-specific AI procurement teams with solutions offices that combine technical specialists, procurement officers, and program managers
- Developing standardized procurement language and templates for AI capabilities that focus on performance metrics rather than technical specifications
- Implementing mandatory cross-agency knowledge sharing of AI implementation successes and lessons learned

Technical Infrastructure for Operational Efficiency

AI implementations will require compute infrastructures across the Federal Government that far exceed the computational capabilities currently available. However, AI innovations extend far beyond user-facing insights and efficiencies, and in many cases AI-based solutions can close the computational gaps that may be created by model implementation. To ensure AI solutions are implemented on a near turn-key basis within the Government, Federal agencies must seek not only front-facing AI solutions, but also backend processing innovations.

Addressing Implementation Barriers

Several barriers currently impede effective Government AI adoption and must be addressed systematically. Legacy system integration requirements significantly complicate deployment of advanced AI capabilities, often forcing innovative solutions to be watered down to accommodate outdated infrastructure. This is compounded by a risk-averse acquisition culture that prioritizes compliance over capability, creating evaluation frameworks that favor documented specifications rather than demonstrated

performance. Furthermore, limited technical expertise among procurement professionals evaluating AI solutions leads to difficulty distinguishing between genuine innovation and marketing hyperbole. By specifically targeting these barriers through policy reform, training programs, and institutional adjustments, the federal government can dramatically accelerate its ability to harness private sector AI innovations while ensuring taxpayer resources support technologies that deliver measurable operational improvements.

Transformational Efficiency Through AI Implementation

The transformative potential of AI in Government operations extends far beyond incremental improvements. When properly evaluated and deployed, AI systems can deliver order-of-magnitude efficiency gains across federal functions.

Current Government operations frequently involve substantial manual processing, resource-intensive computational tasks, and complex decision-making processes with limited technological support. The strategic implementation of AI can fundamentally transform these workflows by:

- Accelerating processing speeds while maintaining or improving accuracy, allowing agencies to clear backlogs and respond more rapidly to emerging situations
- Dramatically reducing infrastructure requirements through more efficient computational approaches
- Enabling previously impossible capabilities by automating complex cognitive tasks that were previously beyond the scope of technological assistance

Measuring Transformation: Efficiency Multipliers

To realize similar efficiency multipliers across government operations, federal agencies should implement a consistent framework for evaluating AI solutions based on quantifiable efficiency metrics:

1. **Baseline Current Performance:** Prior to AI implementation, agencies should document existing process throughput, accuracy, resource requirements, and operational limitations to establish clear performance baselines.

2. **Define Comprehensive Metrics:** Agencies should develop metrics that capture both performance improvements (e.g., speed, accuracy, capability) and resource reductions (e.g., infrastructure, energy, personnel time) to calculate true efficiency multipliers.
3. **Require Quantifiable Demonstrations:** Vendor demonstrations should provide concrete evidence of efficiency gains through live system operation with representative agency data and workflows.
4. **Calculate Total Efficiency Impact:** Evaluation should consider the combined effect of performance improvements and resource reductions to determine the total efficiency multiplier offered by proposed AI solutions.
5. **Implement Verification Procedures:** Post-deployment verification should confirm that promised efficiency gains are realized in actual operations, with contractual remedies if targets are not achieved.

Case Study: WindBorne's AI Weather Intelligence Systems

WindBorne Systems demonstrates how private sector AI solutions can deliver transformational efficiency improvements to Government operations using our advanced weather intelligence platforms. These systems exemplify the principles advocated throughout this response and stand ready for rigorous demonstration and evaluation.

WeatherMesh: AI-Based Numerical Weather Prediction

WindBorne's WeatherMesh fundamentally transforms weather forecasting through advanced AI techniques. This system consistently outperforms traditional and AI-based global models on multiple benchmarks—including those from Google, Huawei, ECMWF, and NOAA's GFS—while requiring a fraction of the computational resources.

The computational efficiency gains achieved by WeatherMesh exemplify the order-of-magnitude improvements possible through AI implementation. Traditional physics-based forecasting systems require massive supercomputing infrastructure costing hundreds of millions, or billions, in infrastructure and millions in annual energy consumption. In stark contrast, WeatherMesh executes a complete 10-day global forecast in under 60 seconds on a single GPU—representing a computational efficiency improvement exceeding 140,000x compared to traditional approaches. This dramatic reduction in computational requirements doesn't come at the cost of

accuracy; instead, WeatherMesh delivers comparable or superior forecast quality while fundamentally transforming the economics and accessibility of advanced weather prediction capabilities.

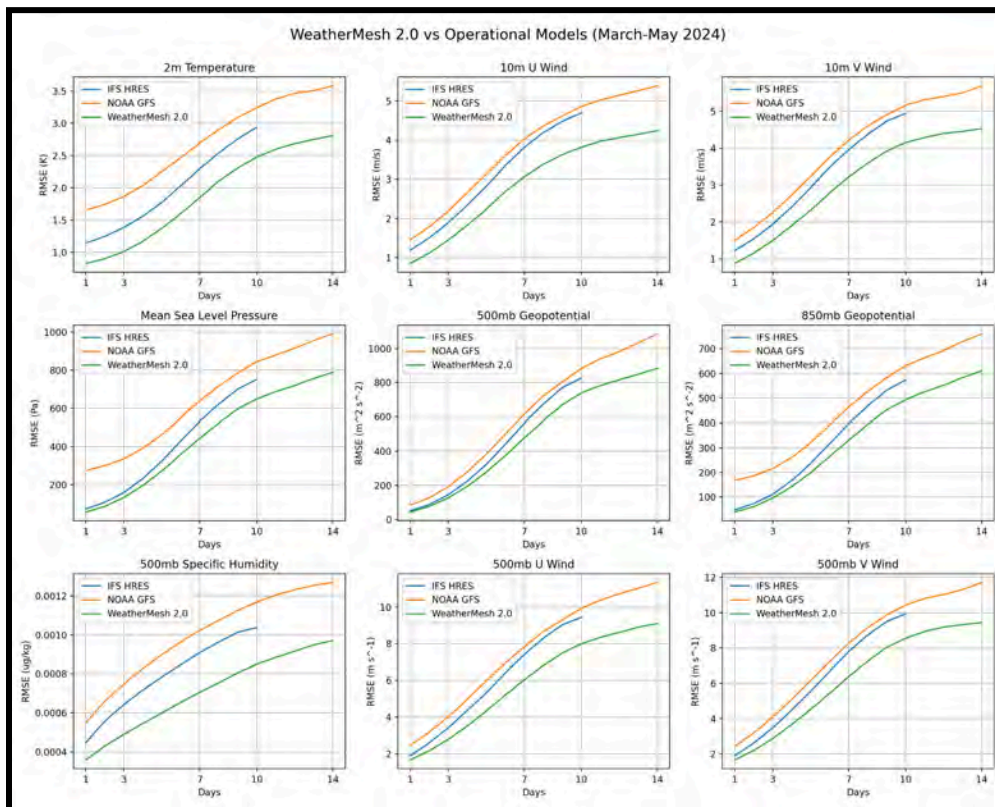


Figure 01: WindBorne's artificial intelligence numerical weather prediction (AIWP) model outperforms GFS and IFS in RMSE across all the above variables, where the WeatherMesh margin for error is represented in green.

WeatherMesh provides ensemble model forecasts on a 0.25-degree global grid for 14+ days, predicting surface and atmospheric variables including temperature, wind, precipitation, and cloud cover at arbitrary resolutions.

WindBorne Catalyst: Infrastructure Optimization

WindBorne's Catalyst system demonstrates how AI can simultaneously improve performance while reducing resource requirements. This integrated hardware and software solution provides continuous forecasting with new global predictions every 10 minutes, rather than the traditional 6-hour cycle, achieving a 36x improvement in update frequency. It features a reduced infrastructure footprint to approximately half of

a standard 42U server rack, replacing the room-sized traditional installations that typically house conventional forecasting systems.

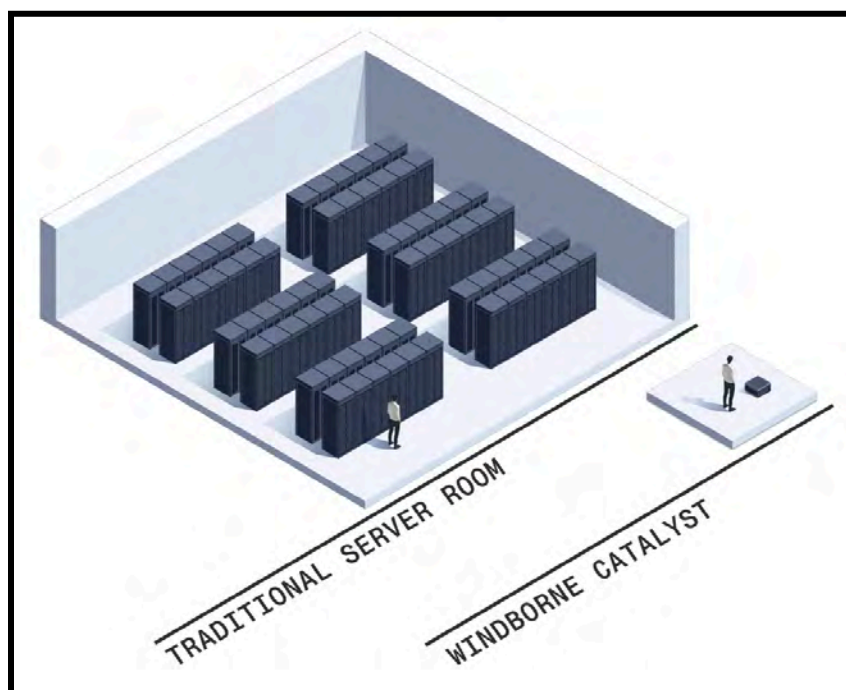


Figure 02: Catalyst can be implemented with WeatherMesh in tandem or as a standable hardware solution to augment traditional NWP systems.

Additionally, Catalyst achieves dramatically lower energy consumption compared to traditional forecasting systems, supporting Federal sustainability goals while delivering superior performance. Procuring WindBorne Catalyst carries approximately 1/100th of the cost of super computers used to support existing Federal weather forecasting operations. This implementation demonstrates how properly selected AI solutions can transform Government operations while simultaneously reducing costs and environmental impact.

Alignment with National Security and Economic Leadership

This procurement-centered approach directly enhances America's national security posture by accelerating deployment of advanced capabilities to defense and intelligence agencies, closing capability gaps with potential adversaries. It enables more rapid identification of adversarial AI capabilities through continuous performance evaluation in realistic scenarios. The approach strengthens industrial base resilience

through diversified AI suppliers and consistent assessment standards, while ensuring security requirements are integrated into AI procurement from the earliest stages, and not added as afterthoughts.

Simultaneously, it drives economic competitiveness by creating clear market incentives for private sector AI development aligned with national priorities. The framework establishes objective performance standards that reward genuine innovation rather than marketing claims, supporting American AI companies competing in global markets through references and proven performance metrics. By accelerating beneficial AI applications across government services, this approach improves citizen experiences while demonstrating American technological leadership in practical applications that can be emulated worldwide.

Conclusion

The artificial intelligence revolution represents both the greatest economic opportunity and the most significant competitive challenge America has faced in a generation. The Federal Government's response will fundamentally shape whether the United States maintains its position as the world's leading technological innovator or cedes this advantage to strategic competitors.

The Federal Government's AI strategy must recognize market realities while leveraging America's unique advantages. By positioning Government agencies as sophisticated customers rather than parallel developers, and by implementing rigorous demonstration-based procurement processes, we can ensure taxpayer resources support genuinely effective technologies that deliver measurable improvements to government operations.

Through implementation of these principles across procurement, infrastructure, workforce, and international domains, the Administration has an unprecedented opportunity to accelerate American AI leadership. A forward-looking approach that harnesses private sector innovation, while addressing legitimate security concerns, will position the United States to lead not just in AI development but in the broader technological revolution it enables.

The window for securing America's AI leadership is narrow but still open. With decisive action guided by market realities rather than bureaucratic imperatives, this Administration can ensure that artificial intelligence advances American prosperity,

security, and values for generations to come and that the United States Government maintains global leadership in technological innovation.

*WindBorne Systems is a Palo Alto-based company pioneering advanced AI systems for weather intelligence and atmospheric monitoring. Our platforms include **WeatherMesh**, the world's leading AI-based numerical weather prediction model, and our AI-enabled Global Sounding Balloon constellation, **Atlas**, for atmospheric data collection. This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.*